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Constraints on the cosmic expansion history from GWTC-3

Authors: Abbott et al. (LIGO-Virgo-KAGRA)

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ABSTRACT

Uses 42 BBH events SNR>11 from GWTC-3 with GLADE+ catalog (20% complete to 800 Mpc). Tests flat w0CDM and flat LCDM with priors H0 uniform [10,200] or [65,77] etc. Dominated by population assumptions. Joint with GW170817 bright siren. Shows dependence on galaxy weighting (g-band luminosity vs uniform) and cosmology prior.

What was actually measured (numbers):

GLADE+ 20% complete 800 Mpc, H0 prior [10,200], w0 prior [-3,0], Omega_m uniform [0,1]

Devices / Protocol:

LIGO-Virgo-KAGRA O3, GLADE+

Replication across teams / Independence:

Second-generation GW catalog confirms earlier O1/O2 result with larger sample

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